

Galois Theory / MM7043 2026W

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1 Fields

Throughout the semester, we will employ the following conventions:

- Unless otherwise specified, all rings will be assumed to be unital and commutative.
- A *ring homomorphism* is a map of rings $\varphi : R \rightarrow S$, such that
 1. $\varphi(x + y) = \varphi(x) + \varphi(y)$, for all $x, y \in R$.
 2. $\varphi(x \cdot y) = \varphi(x) \cdot \varphi(y)$, for all $x, y \in R$.
 3. $\varphi(1_R) = 1_S$

Definition 1.1. Let R be a ring. An element $x \in R$ is called a *unit* or *invertible*, if there exists an element $y \in R$ such that $x \cdot y = 1$.

The set of all units of R form a group under multiplication and will be denoted R^* .

Example 1.2.

1. $\mathbb{Z}^* = \{-1, 1\}$
2. $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$
3. For $n \in \mathbb{N}$, it holds that $(\mathbb{Z}_n)^* = \{m \in \mathbb{Z}_n : \gcd(m, n) = 1\}$.

Definition 1.3. A *field* is a ring $\mathbb{K}$ such that $\mathbb{K}^* = \mathbb{K} \setminus \{0\}$.

Example 1.4.

1. $\mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$
2. $\mathbb{Z}_p$, where $p \in \mathbb{P}$.

Lemma 1.5. *A ring R is a field if and only if the only ideals of R are $\{0\}$ and R .*

Proof. Let $I \subseteq R$ be an ideal. If I contains a unit then $I = R$.

“ $\implies$ ”: If $I \subseteq R$ is a non-zero, non-trivial ideal, then I contains a unit, which implies $I = R$.

“ $\impliedby$ ”: If R is not a field, then there exists some non-zero, non-unit element $x \in R$. But then the ideal Rx satisfies

$$\{0\} \subsetneq Rx \subsetneq R. \quad \square$$

Corollary 1.6. *A homomorphism of fields is always injective.*

Proof. Let $\varphi : \mathbb{K} \rightarrow \mathbb{L}$ be a field homomorphism. Then $\ker \varphi$ is an ideal in $\mathbb{K}$. We have $1_{\mathbb{K}} \notin \ker \varphi$, therefore $\ker \varphi = \{0\}$ and φ is injective. $\square$

1.1 Field extensions

We can construct new fields by considering subfields generated by a subset of a field.

Lemma 1.7. *Suppose $\mathbb{K}$ is a field and $\{F_\alpha : \alpha \in I\}$ is a family of subfields of $\mathbb{K}$. Then*

$$\bigcap_{\alpha \in I} F_\alpha$$

is a subfield of $\mathbb{K}$.

Definition 1.8. Suppose $\mathbb{L}_0 \subseteq \mathbb{L}$ are fields and $\alpha_1, \alpha_2, \dots \in \mathbb{L}$. Then we define $\mathbb{L}_0(\alpha_1, \alpha_2, \dots)$ as the intersection of all subfields of $\mathbb{L}$ that contain $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$, and will also be called the *extension* of $\mathbb{L}_0$ generated by $\alpha_1, \alpha_2, \dots$.

It is clear that such an extension is the smallest subfield of $\mathbb{L}$ that contains $\mathbb{L}_0$ and $\alpha_1, \alpha_2, \dots$.

Example 1.9.

1. Consider the fields $\mathbb{Q} \subseteq \mathbb{R}$, then one can show

$$\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}.$$

2. Since the element $1_{\mathbb{Q}}$ already generates $\mathbb{Q}$ as a field, it follows that $\mathbb{Q}$ has no proper subfields, i.e. $\mathbb{Q}$ is a *prime field*.
3. The field $\mathbb{F}_p$ is a primary field for any $p \in \mathbb{P}$.

Definition 1.10. Let $\mathbb{F}$ be a field. The *primary subfield* of $\mathbb{F}$ is the subfield generated by the element $1_{\mathbb{F}}$.

Definition 1.11. Let R be a ring, then its *characteristic* is defined by

$$\text{char } R := \begin{cases} \min\{n \in \mathbb{N} : \sum_{j=1}^n 1_R = 0\}, & \text{if it exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 1.12. Let $\mathbb{F}$ be a field.

- If $\text{char } \mathbb{F} = 0$, then the primary subfield of $\mathbb{F}$ is¹ $\mathbb{Q}$.
- If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then the primary subfield of $\mathbb{F}$ is $\mathbb{F}_p$.

Lemma 1.13. Suppose $M \subseteq R$ is a maximal ideal. Then the quotient R/M is a field.

Remark 1.14. Suppose $\mathbb{K}$ is a field and consider $\mathbb{K}[x]$, the ring of polynomials in x with coefficients in $\mathbb{K}$. Recall that $\mathbb{K}[x]$ is a principle ideal domain. A non-trivial ideal $(f(x))$ is maximal in $\mathbb{K}[x]$ if and only if $f(x)$ is an irreducible polynomial. In particular, $\mathbb{K}[x]/(f(x))$ is a field whenever $f(x)$ is irreducible over $\mathbb{K}$.

Moreover, $\mathbb{K}[x]/(f(x))$ contains $\mathbb{K}$ as a subfield, via the homomorphisms

$$\mathbb{K} \hookrightarrow \mathbb{K}[x] \rightarrow \mathbb{K}[x]/(f(x)).$$

Therefore, for every irreducible $f(x)$, $\mathbb{K}[x]/(f(x))$ is a field extension of $\mathbb{K}$.

Note that $f(x)$ being irreducible with $\deg f > 1$ implies that $f(x)$ has no roots in $\mathbb{K}$. But $f(x)$ always has a root in $\mathbb{K}[x]/(f(x))$, because $f(x)$ is zero as an element of $\mathbb{K}[x]/(f(x))$.

The above demonstrates (roughly) that there are relationships:

$$\begin{array}{ll} \text{Polynomials} & \longleftrightarrow \text{Field extensions} \\ \text{Properties of roots} & \longleftrightarrow \text{Symmetries of field extensions} \end{array}$$

Practically, we want some criteria to determine if a polynomial is irreducible.

¹We always mean “up to isomorphism”.

Remark 1.15. If $\mathbb{K}$ is a field and $p(x) \in \mathbb{K}[x]$, $\deg p > 1$ is a polynomial with a root in $\mathbb{K}$, then p is not irreducible. Indeed, for any $a \in \mathbb{K}$ one can write

$$p(x) = q(x)(x - a) + p(a),$$

for some polynomial $q(x) \in \mathbb{K}[x]$. If a is a root of $p(x)$, this demonstrates that $p(x)$ is not irreducible.

Note that the converse fails: The polynomial $(x^2 + 1)^2$ has no roots in $\mathbb{R}$ and is not irreducible.

Lemma 1.16. *For a field $\mathbb{K}$, a polynomial $p(x) \in \mathbb{K}[x]$ such that $2 \leq \deg p \leq 3$, is irreducible if and only if it has no roots in $\mathbb{K}$.*

Lemma 1.17 (Gauss). *Suppose $p(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ is primitive, that is $\gcd(a_0, a_1, \dots, a_n) = 1$. Then $p(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.*

Lemma 1.18 (Eisenstein's Criterion). *Let $q(x) = a_0 + a_1x + \dots + a_nx^n \in \mathbb{Z}[x]$ and suppose there exists a prime $p \in \mathbb{P}$ such that:*

1. $p \nmid a_n$
2. $p \mid a_0, \dots, a_{n-1}$
3. $p^2 \nmid a_0$

Then $q(x)$ is irreducible in $\mathbb{Q}[x]$.

Example 1.19.

1. The polynomial

$$q(x) = 2x^3 + 6x^2 - 9x + 15$$

satisfies Eisenstein's criterion with $p = 3$.

2. The polynomial

$$q(x) = x^2 + x + 1$$

does not satisfy Eisenstein's criterion. Nevertheless, one can consider the transformation

$$r(x) = q(x + 1) = (x + 1)^2 + (x + 1) + 1 = x^2 + 3x + 3,$$

which does satisfy the criterion. Hence $q(x)$ is also irreducible.

More generally, using the same method as above, one can show that for any prime $p \in \mathbb{P}$, the polynomial

$$q(x) = x^{p-1} + x^{p-2} + \dots + 1$$

is irreducible.

Example 1.20. We recall the following basic homomorphisms:

1. Suppose R is a ring. Then there exists a unique ring homomorphism

$$\varepsilon : \mathbb{Z} \rightarrow R, n \mapsto n \cdot 1_R.$$

2. Suppose $\mathbb{F}$ is a field. Let $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ be the ring homomorphism from the above and let $I = \ker \varepsilon$. Then ε induces a monomorphism $\mathbb{Z}/I \rightarrow \mathbb{F}$. In particular, $\mathbb{Z}/I$ is an integral domain, therefore $I \subseteq \mathbb{Z}$ is a prime ideal. Thus $I = \{0\}$, or $I = (p)$ for some $p \in \mathbb{P}$. This “formalizes” our previous definition of $\text{char } \mathbb{F}$.

If $\text{char } \mathbb{F} = 0$, then $\varepsilon : \mathbb{Z} \rightarrow \mathbb{F}$ is a monomorphism, so ε extends to a monomorphism $\mathbb{Q} \rightarrow \mathbb{F}$.

If $\text{char } \mathbb{F} = p \in \mathbb{P}$, then $\text{ran } \varepsilon \cong \mathbb{Z}/p\mathbb{Z} \cong \mathbb{F}_p$.

3. Suppose we have a fixed homomorphism of rings $i : R \rightarrow S$. Then for every choice of $\theta \in S$ there exists a unique ring homomorphism

$$\text{ev}_\theta : R[x] \rightarrow S,$$

such that $\text{ev}_\theta(r) = i(r)$, for all $r \in R$ and $\text{ev}_\theta(X) = \theta$, namely the map $p(x) \mapsto p(\theta)$.

Definition 1.21. Let $\mathbb{F}$ be a field. A (*field*) *extension* of $\mathbb{F}$ is a field $\mathbb{K}$ containing $\mathbb{F}$ as a subfield². We will also write $\mathbb{F} \hookrightarrow \mathbb{K}$ to denote such an extension.

Example 1.22.

1. $\mathbb{Q} \hookrightarrow \mathbb{R}$
2. $\mathbb{R} \hookrightarrow \mathbb{C}$

Remark 1.23. Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension. Then $\mathbb{K}$ is a vector space over $\mathbb{F}$.

Definition 1.24. The *degree* of a field extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is defined as the dimension of $\mathbb{K}$ as a vector space over $\mathbb{F}$. We also denote this by $[\mathbb{K} : \mathbb{F}]$ or $\dim_{\mathbb{F}} \mathbb{K}$.

Example 1.25.

1. $[\mathbb{R} : \mathbb{Q}] = \infty$
2. $[\mathbb{C} : \mathbb{R}] = 2$

²Again, we mean “containing” up to a monomorphism.

Lemma 1.26. Suppose we have field extensions $\mathbb{F} \hookrightarrow \mathbb{K} \hookrightarrow \mathbb{L}$, then

$$[\mathbb{L} : \mathbb{F}] = [\mathbb{L} : \mathbb{K}] \cdot [\mathbb{K} : \mathbb{F}].$$

Definition 1.27. Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension and $\alpha \in \mathbb{K}$. We say that α is *algebraic* over $\mathbb{F}$ if there exists a non-zero polynomial $p(x) \in \mathbb{F}[x]$ such that $p(\alpha) = 0$. Otherwise α is called *transcendental* over $\mathbb{F}$.

Example 1.28.

1. $\sqrt{2}, \sqrt[3]{2}$ are algebraic over $\mathbb{Q}$.
2. π, e are transcendental over $\mathbb{Q}$.

Definition 1.29. We say that an extension $\mathbb{F} \hookrightarrow \mathbb{K}$ is algebraic, if every element $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Otherwise, we say that the extension is transcendental.

1.2 Minimal polynomials

Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ is a field extension and $\alpha \in \mathbb{K}$ is an algebraic element over $\mathbb{F}$. Then

$$I = \{p(x) \in \mathbb{F}[x] : p(\alpha) = 0\},$$

is a nonzero ideal in $\mathbb{F}[x]$. Since $\mathbb{F}[x]$ is a principle ideal domain, there exists a unique monic polynomial $m_{\alpha, \mathbb{F}}(x) \in \mathbb{F}[x]$, such that $I = (m_{\alpha, \mathbb{F}})$. If $\mathbb{F}$ is clear from the context, we also abbreviate this as m_α .

Definition 1.30. The polynomial $m_{\alpha, \mathbb{F}}(x)$ in the is called the *minimal polynomial* of α .

Lemma 1.31. A monic polynomial $p(x) \in \mathbb{F}[x]$ is a minimal polynomial of $\alpha \in \mathbb{K}$ if and only if

1. $p(\alpha) = 0$, and
2. $p(x)$ is irreducible over $\mathbb{F}$.

Proof. “ $\implies$ ”: Suppose $p(x) = m_\alpha(x)$. By definition, $p(\alpha) = 0$ and $p(x)$ generates the ideal of polynomials vanishing at α . Suppose $p(x) = q_1(x) \cdot q_2(x)$. Then $p(\alpha) = q_1(\alpha) \cdot q_2(\alpha)$. Without loss of generality, assume that $q_1(\alpha) = 0$, that is $q_1(x) \in (p(x))$, therefore $p(x) \mid q_1(x)$. But also, $q_1(x) \mid p(x)$ by our assumption. Therefore $p(x)$ and $q_1(x)$ are associate in $\mathbb{F}[x]$, hence $q_2(x)$ is a unit.

“ $\impliedby$ ”: Similarly to the above. □

Suppose $F \hookrightarrow \mathbb{K}$ and $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$. Since $m_{\alpha, \mathbb{F}}(x)$ is irreducible, $(m_{\alpha, \mathbb{F}})$ is maximal, hence $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x))$ is a field.

Proposition 1.32. *In the context above, $\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) \cong \mathbb{F}(\alpha)$.*

Proof. Consider the evaluation homomorphism

$$\text{ev}_\alpha : \mathbb{F}[x] \rightarrow \mathbb{K}.$$

The image of ev_α is $\mathbb{F}[\alpha]$, the subring of $\mathbb{K}$ generated by $\mathbb{F}$ and α . But the image is also isomorphic to

$$\mathbb{F}[x]/(m_{\alpha, \mathbb{F}}(x)) = \mathbb{F}[x]/\ker \text{ev}_\alpha,$$

which is a field. □

Corollary 1.33. *In the context above, $[\mathbb{F}(\alpha) : \mathbb{F}] = \deg m_{\alpha, \mathbb{F}}$.*

Proof. This follows from the general fact that, for a monic polynomial $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$, it holds that

$$\mathbb{F}[x]/(p(x)) \cong \{b_0 + b_1x + \dots + b_{n-1}x^{n-1} : b_0, \dots, b_{n-1} \in \mathbb{F}\} \cong \mathbb{F}^n. \quad \square$$

Note that conversely, if $p(x) = a_0 + \dots + a_{n-1}x^{n-1} + x^n \in \mathbb{F}[x]$ is an irreducible polynomial, then $\mathbb{F}[x]/(p(x))$ is a field extension of $\mathbb{F}$ of degree n . Furthermore, the element $x + (p(x)) \in \mathbb{F}[x]/(p(x))$ is a zero of $p(x)$, and $p(x)$ is the minimal polynomial of $x + (p(x))$.

Lemma 1.34. *Let $\mathbb{F} \hookrightarrow \mathbb{K}$ be a field extension, then $\alpha \in \mathbb{K}$ is algebraic over $\mathbb{F}$ if and only if $[\mathbb{F}(\alpha) : \mathbb{F}] < \infty$.*

Proof. “ $\implies$ ”: Follows immediately from the previous corollary.

“ $\impliedby$ ”: The elements $1, \alpha, \alpha^2, \dots, \alpha^n$ are linearly dependant over $\mathbb{F}$. Thus there exists $s_0, s_1, \dots, s_n \in \mathbb{F}$, not all zero, such that $s_0 + s_1\alpha + \dots + s_n\alpha^n = 0$, therefore α is algebraic over $\mathbb{F}$. □

Corollary 1.35. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$ and $\alpha, \beta \in \mathbb{K}$ are algebraic over $\mathbb{F}$. Then $\alpha \pm \beta, \alpha\beta, \frac{1}{\alpha}$ (provided $\alpha \neq 0$) are algebraic over $\mathbb{F}$.*

Proof. By the previous lemma, it suffices to show that the respective generated extensions are of finite degree. But $\mathbb{F}(\alpha \pm \beta), \mathbb{F}(\alpha\beta)$ are subfields of $\mathbb{F}(\alpha, \beta)$. But the latter extension has finite degree, since

$$[\mathbb{F}(\alpha, \beta) : \mathbb{F}] = [\mathbb{F}(\alpha, \beta) : \mathbb{F}(\alpha)] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] \leq [\mathbb{F}(\beta) : \mathbb{F}] \cdot [\mathbb{F}(\alpha) : \mathbb{F}] < \infty.$$

The final claim follows from the fact that $\mathbb{F}(\alpha) = \mathbb{F}(\frac{1}{\alpha})$. □

Corollary 1.36. *Suppose $\mathbb{F} \hookrightarrow \mathbb{K}$. Then the set of all algebraic elements of $\mathbb{K}$ over $\mathbb{F}$ is a sub-extension of $\mathbb{K}$, that is it is a sub-field of $\mathbb{K}$ that is also an extension of $\mathbb{F}$. This is also called the algebraic closure of $\mathbb{F}$ in $\mathbb{K}$.*

Example 1.37. We want to determine the minimal polynomial of $\sqrt{2} + \sqrt{3}$. We consider that we ultimately want $x = \sqrt{2} + \sqrt{3}$, squaring yields $x^2 = 5 + 2\sqrt{6}$, that is $x^2 - 5 = 2\sqrt{6}$. Squaring again yields $x^4 - 10x^2 + 1$. It remains to show that this is minimal, which will be the case if one can show

$$[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4.$$